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Department of Digital Systems

Level: Master's Programme of Studies

Network Security

Palo Alto

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09/01/2026

Abstract

Creation of a secure network using Palo Alto Firewall within a VMware environment. System installation and configuration, with implementation of a strict security policy allowing selective access only to Web, SSH, and port 9001 services. Total blockage of all other traffic via a "Deny-All" rule, followed by verification of internal network isolation through connectivity testing.

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Introduction

The process begins by downloading the required .ova (Open Virtual Appliance) files, which are the pre-configured virtual machines that make up the lab:

- Palo Alto Firewall
- Management Client
- DMZ Server
- Virtual Router

To understand the lab's architecture, it is important to clarify the role of each virtual machine and how they interact in the context of network security. The Virtual Router acts as the connection node to the external network, simulating the exit gateway to the internet (Outside Zone) and managing the routing of traffic to and from the outside world. The central pillar of the infrastructure is the Palo Alto Firewall, which sits between all network zones. Its role is to inspect and filter all data traffic, allowing or rejecting connections based on security policies.

Within the network, the Client is placed in the protected zone (Inside Zone) and is used to manage the Firewall, having controlled access outward while remaining protected from inbound external threats. In contrast, the DMZ Server is placed in an isolated zone. Its purpose is to host services that must be accessible from the internet (such as a Web Server or SSH), while ensuring that, if it is compromised, the attacker will not have direct access to the internal network (Client), since traffic between the DMZ and the Inside Zone is strictly controlled by the Firewall.

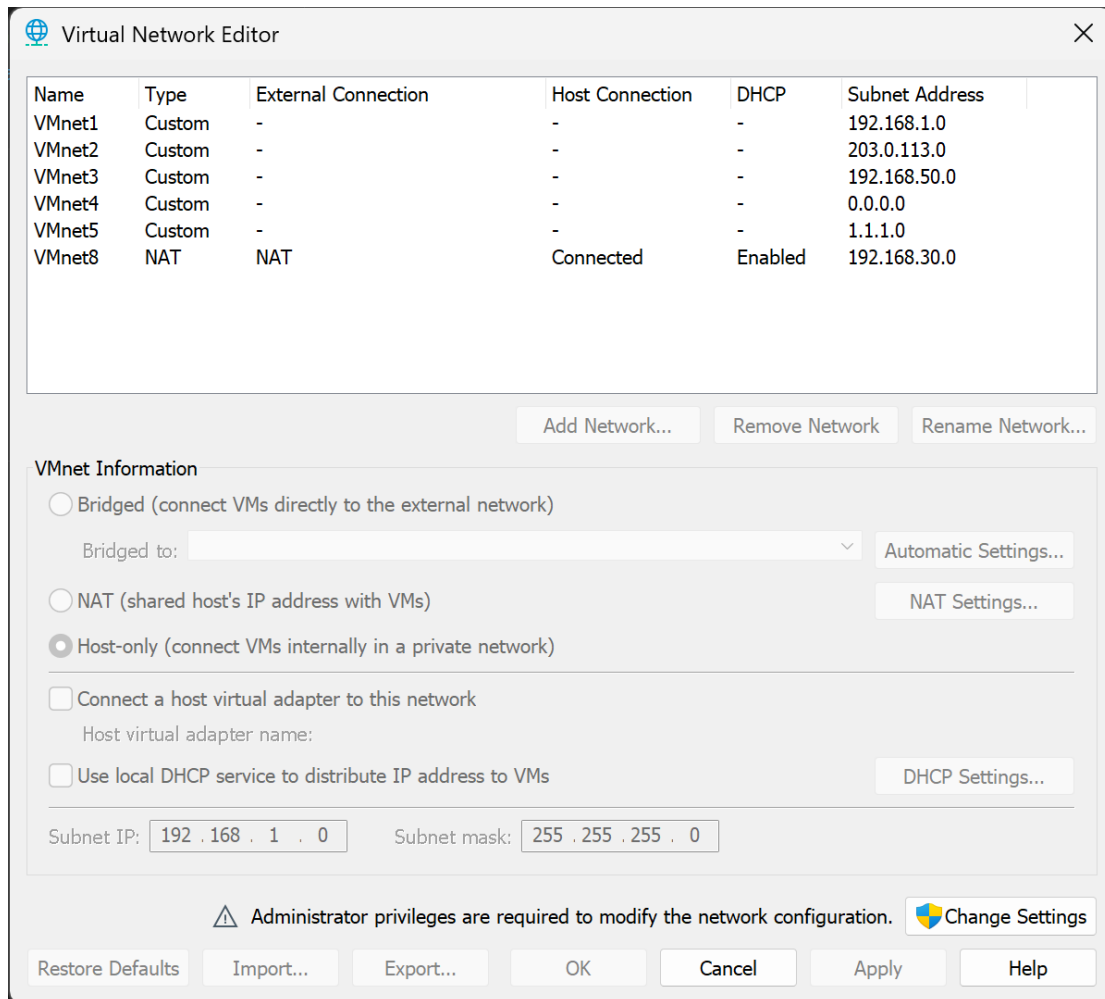
Before starting the machines, the network infrastructure must be configured in VMware Workstation through the Virtual Network Editor. Specifically, the virtual networks (VMnets) are created and configured with the appropriate IP addresses (Subnets), so as to ensure correct traffic routing and communication between the zones (Inside, Outside, DMZ), according to the lab topology.

Once the network is set up, the four virtual machines are imported into VMware. At this stage, a critical action is the manual configuration of the MAC Addresses on each machine's network cards, according to the guide's specifications, in order to avoid conflicts and correctly assign the static IP addresses. Once the hardware settings are

complete, the machines are started in a specific order (VRouter, Firewall, DMZ, Client) to ensure the smooth operation of the network services.

Access to the system is achieved through the Client virtual machine. From the Client environment, using a browser, we connect to the Firewall's management address (192.168.1.254) by entering the administrator credentials. As a final preparation step, the Authorization Code is registered to unlock the necessary functions and receive updates. The lab is now fully operational and ready for the application of the security rules, such as enabling SSH and HTTP to the DMZ and blocking the rest of the traffic, as defined in the assignment brief.

Practical part of the assignment

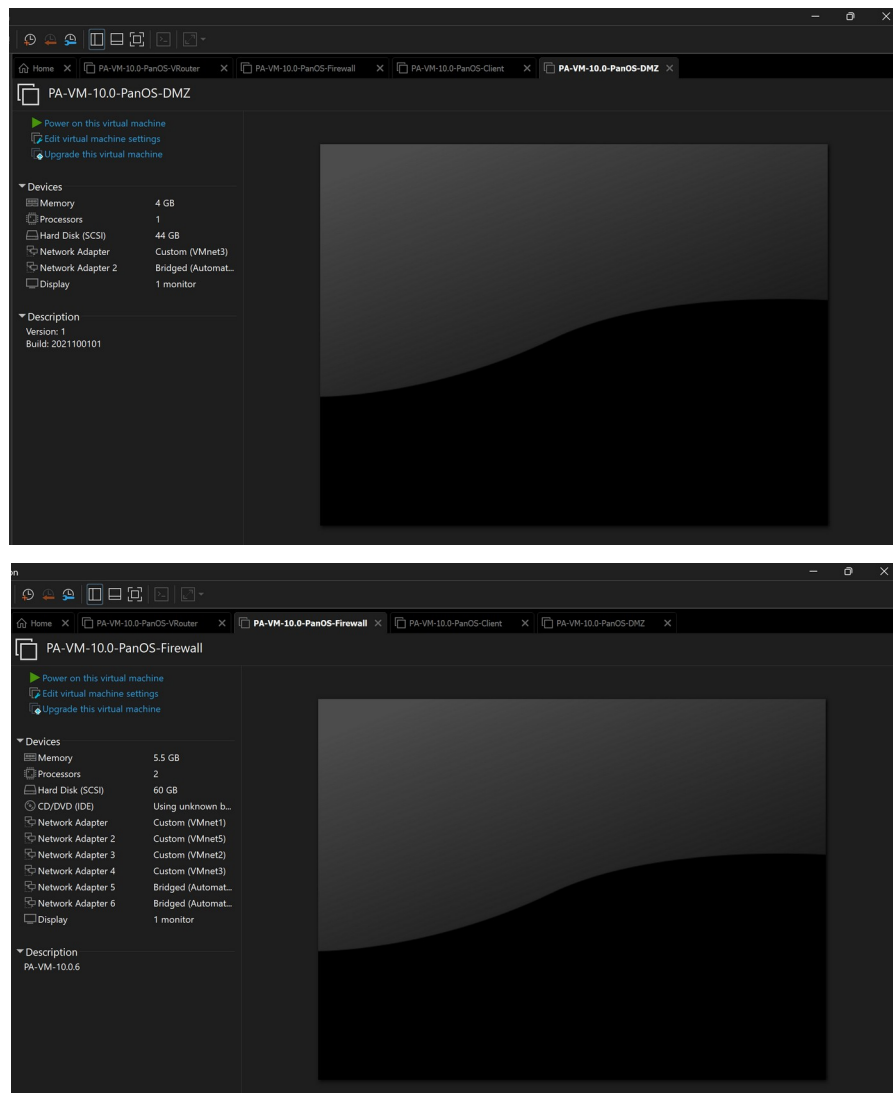


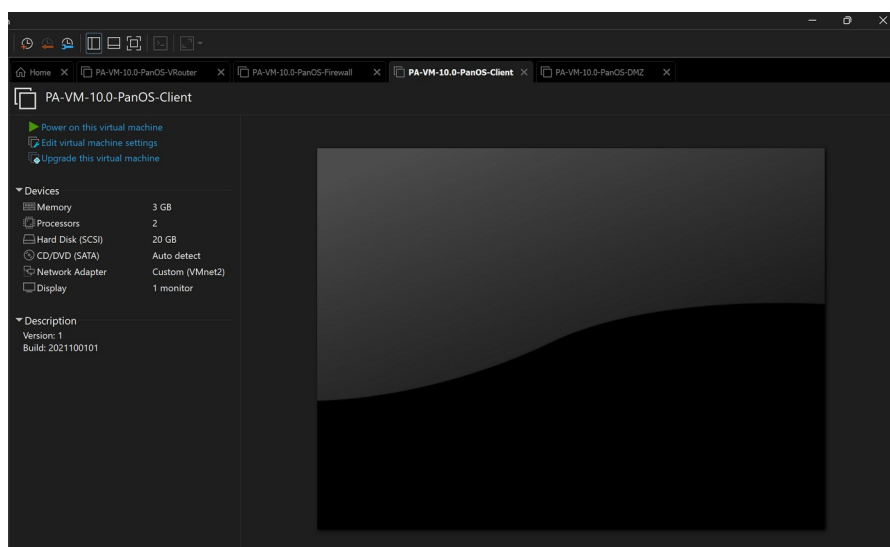
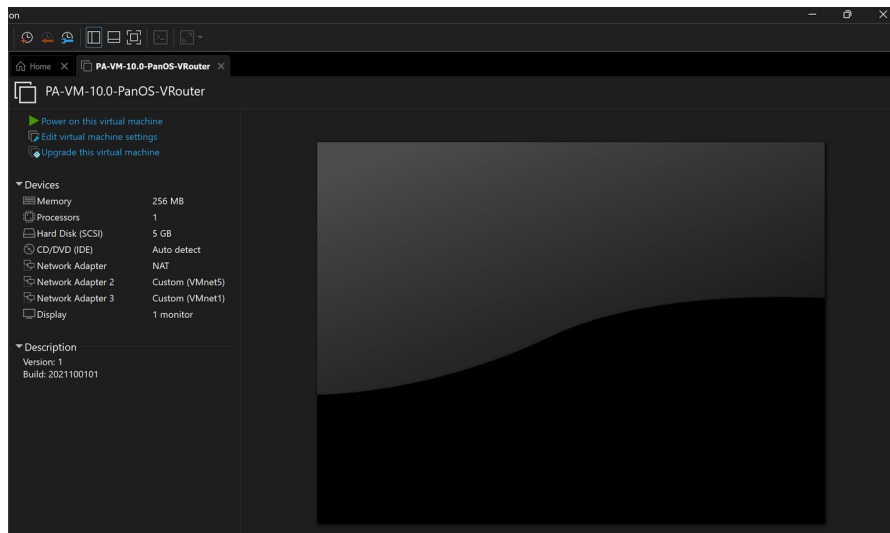
The implementation of the lab began with the configuration of the Virtual Network Editor in VMware Workstation. Operating with administrative privileges, we proceeded to define the Subnet IP addresses, as well as to modify the MAC addresses of the network adapters, so that they match the required topology. We then imported the virtual machines from the provided .ova files, following this startup order:

1. vRouter
2. Firewall (PA-VM)
3. DMZ Server
4. Client

During the process we faced significant delays and technical problems, as repeated loss of internet connectivity (network drop) was observed on the Host computer. The problem was traced to the network card settings, which had initially been set to Bridged mode. This caused an IP Conflict, since the lab's static IPs (192.168.1.x) coincided with our local network's address range, leading to a collapse of the connection. The issue was resolved by faithfully applying the instructions of Palo Alto's official installation guide

(Palo Alto Networks VMware Workstation 10.0 Academy Lab Deployment Guide). Specifically, we set the adapters to Custom mode (VMnets) and NAT for the vRouter, ensuring the isolation of the lab's traffic from the physical network.



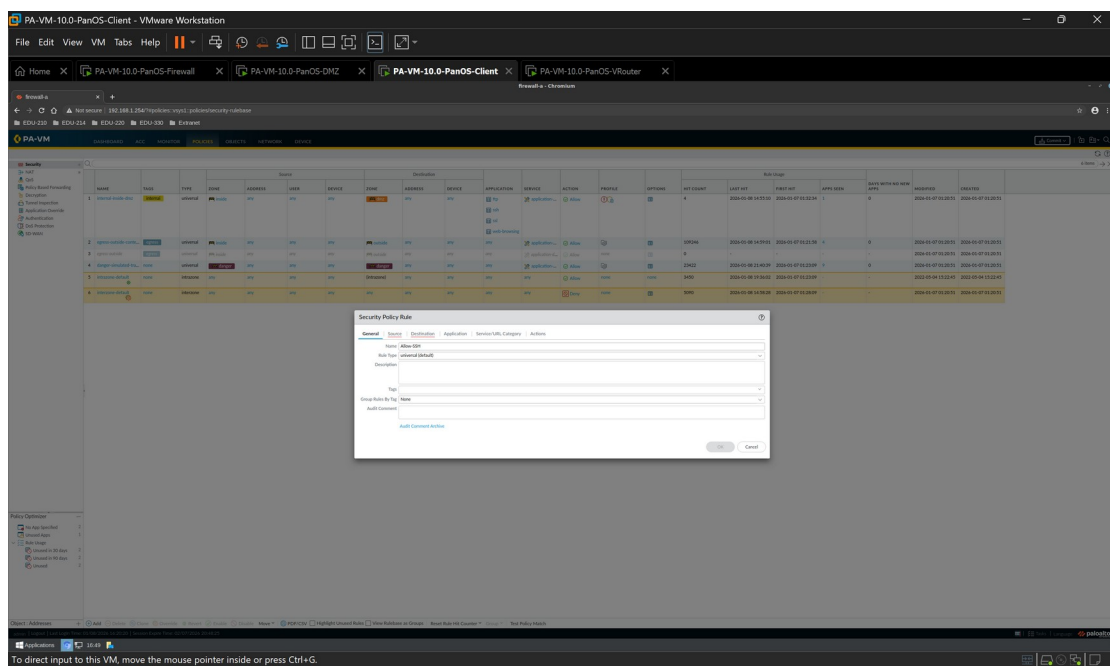
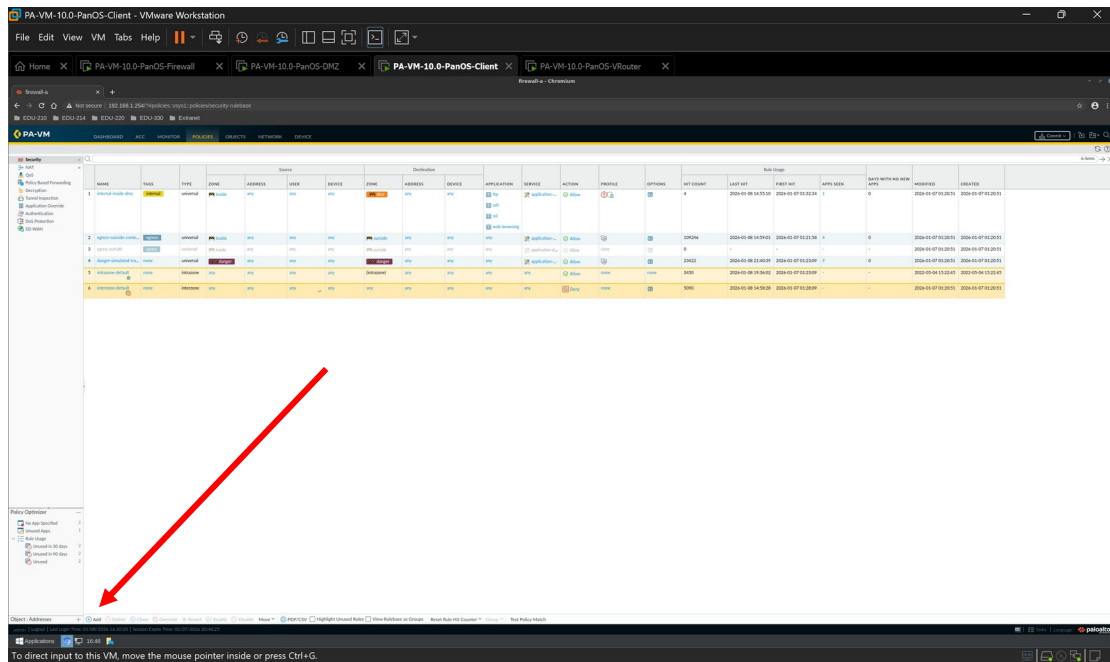


After applying the correct settings, operation was fully restored and the Client started successfully.

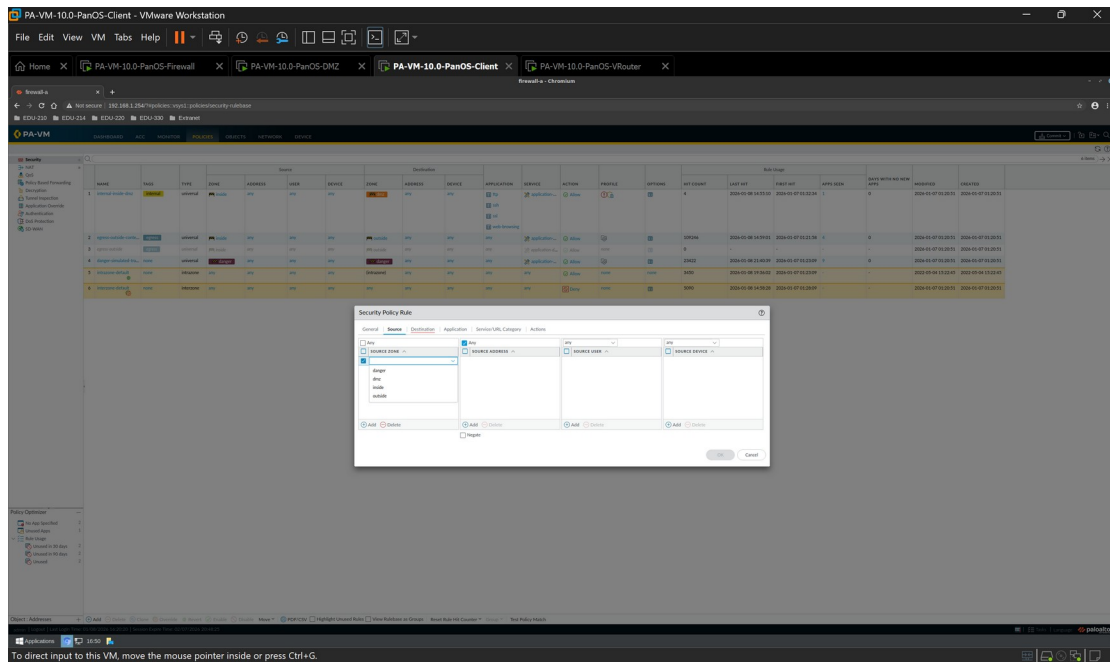
Following the steps of the lab guide, we connected to the management environment of the Palo Alto Firewall. Specifically, we opened the Chromium Web Browser on the Client and entered the address <https://192.168.1.254> to access the login console.

We enter the administrator credentials and then the Web Interface of the Palo Alto Firewall appears.

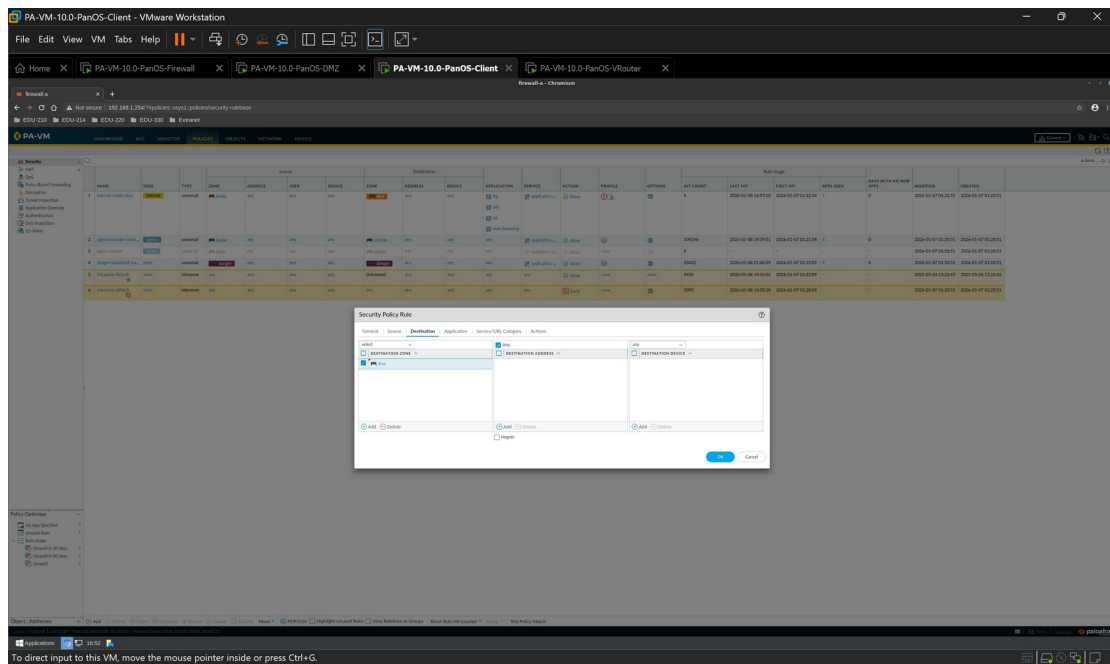
Faithfully following the procedure described in the installation guide, we completed the initial configuration. In the final stage, we went to the Device > Licenses tab and selected 'Activate features using authorization code'. By entering the Authorization Code provided to us, we successfully activated the necessary software Licenses, such as Threat Prevention and DNS Security, which is also confirmed in the screenshot below.



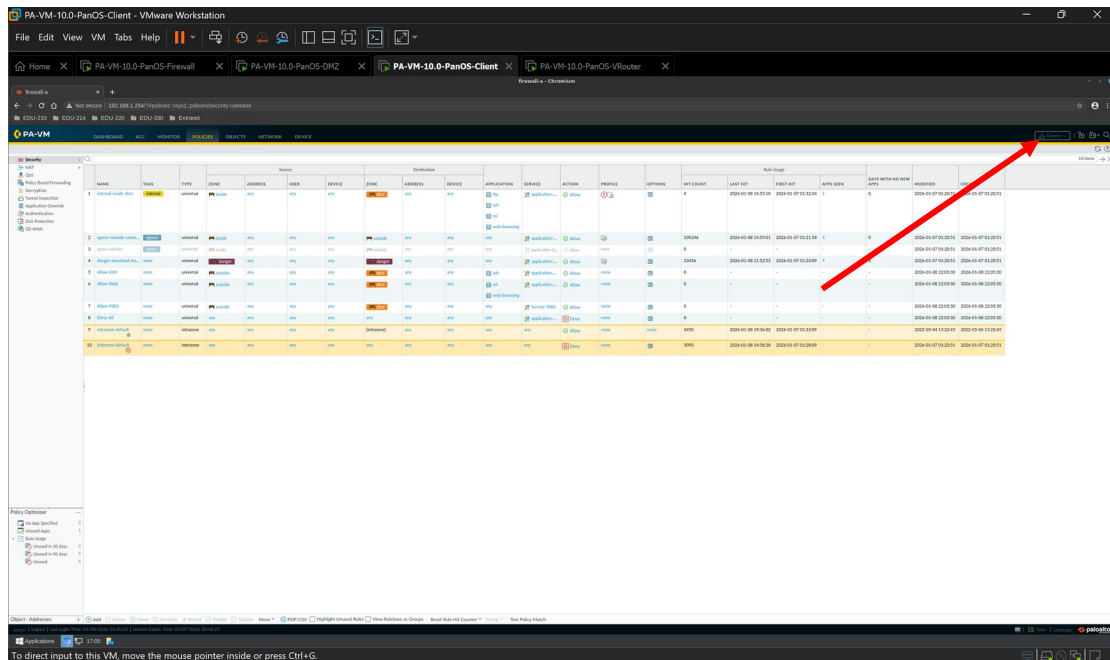
1. In the Source tab, we set the source zone to 'outside', as the traffic comes from the external network.



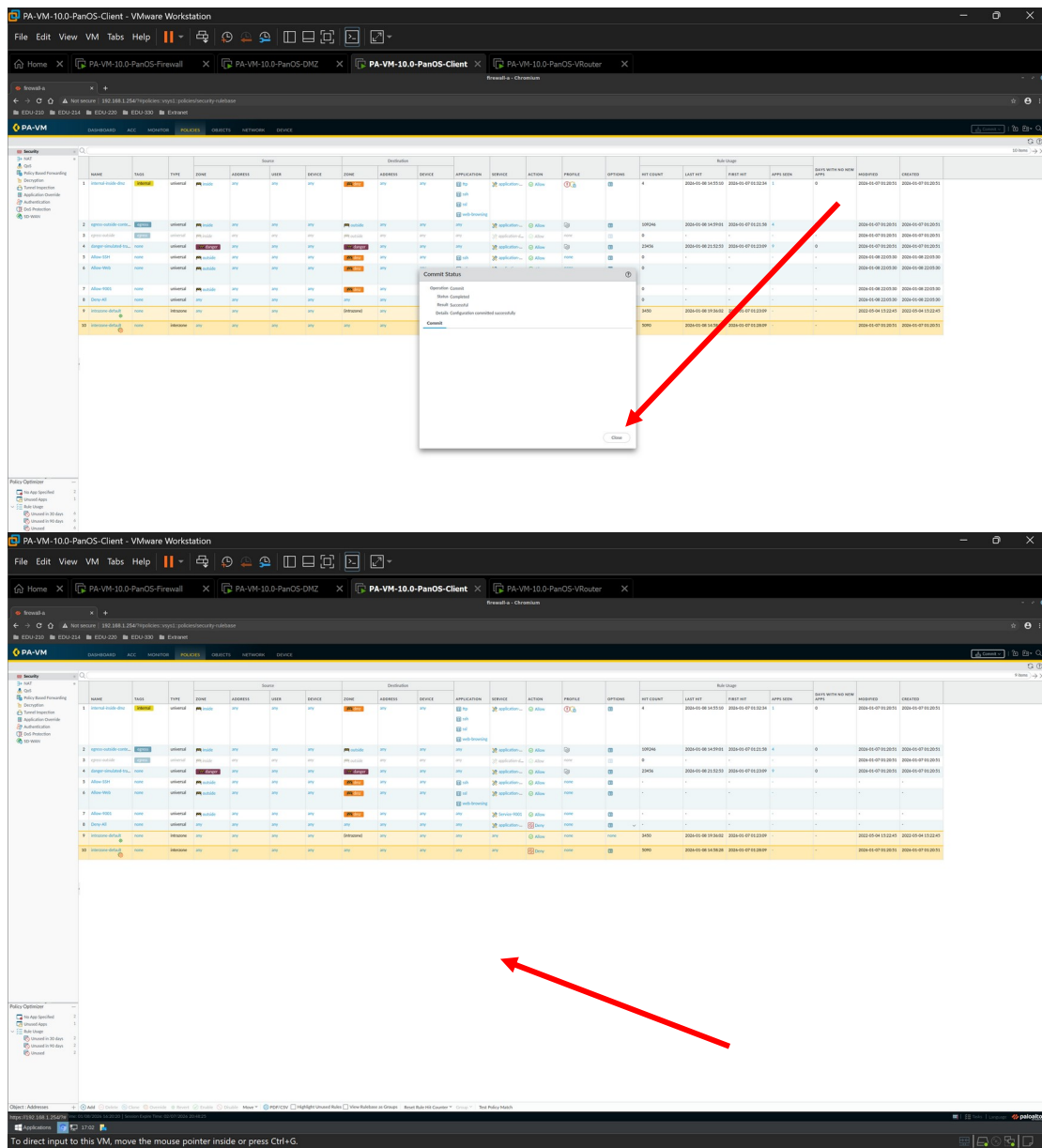
2. In the Destination tab, we selected the 'dmz' zone, where our servers are hosted.



3. In the Application tab, we searched for and selected the 'ssh' application. This allows the Firewall to recognise the SSH protocol regardless of port.

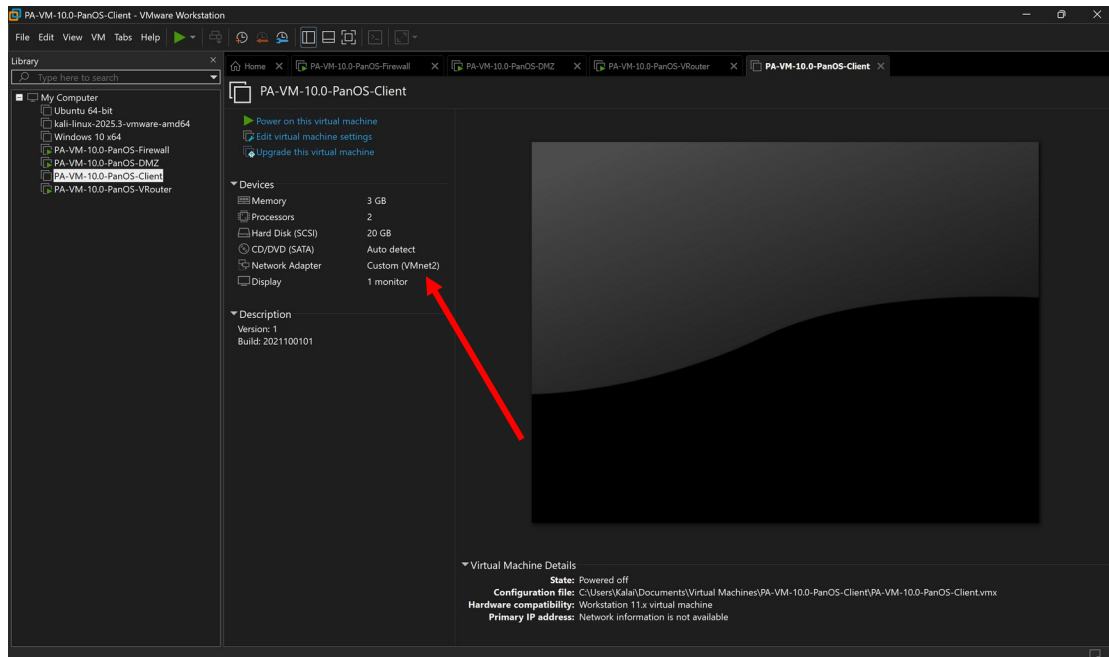


With all four security rules successfully registered, we finalised the settings by selecting Commit. The system confirmed the application of the policy ('Configuration committed successfully').

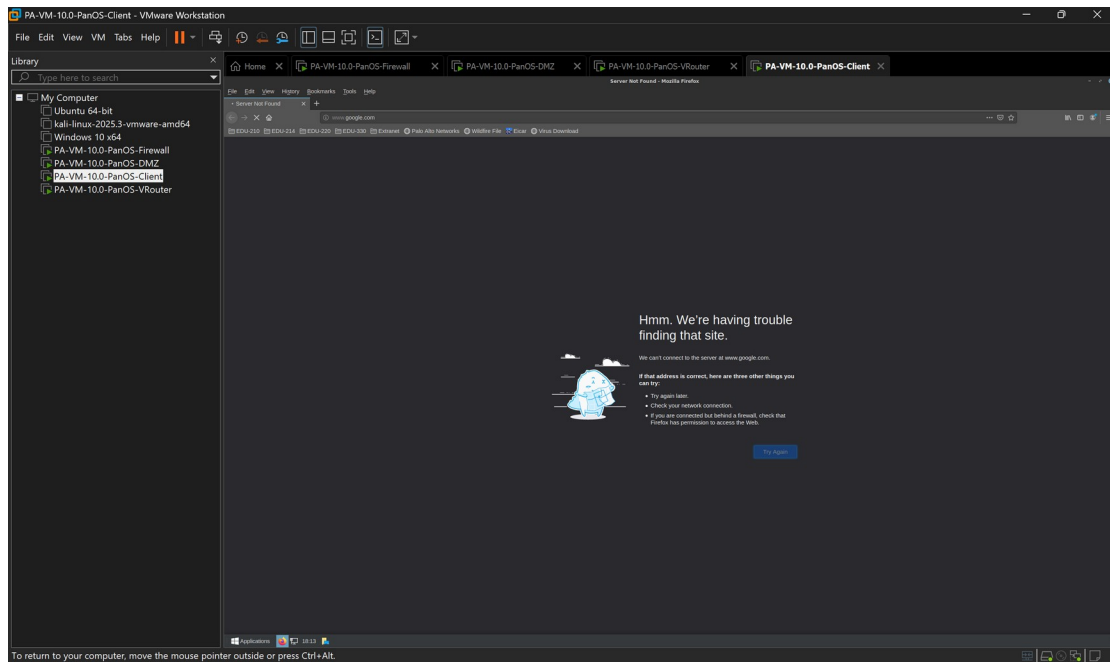


To practically verify the security and correct operation of the rules, we carried out the following test steps:

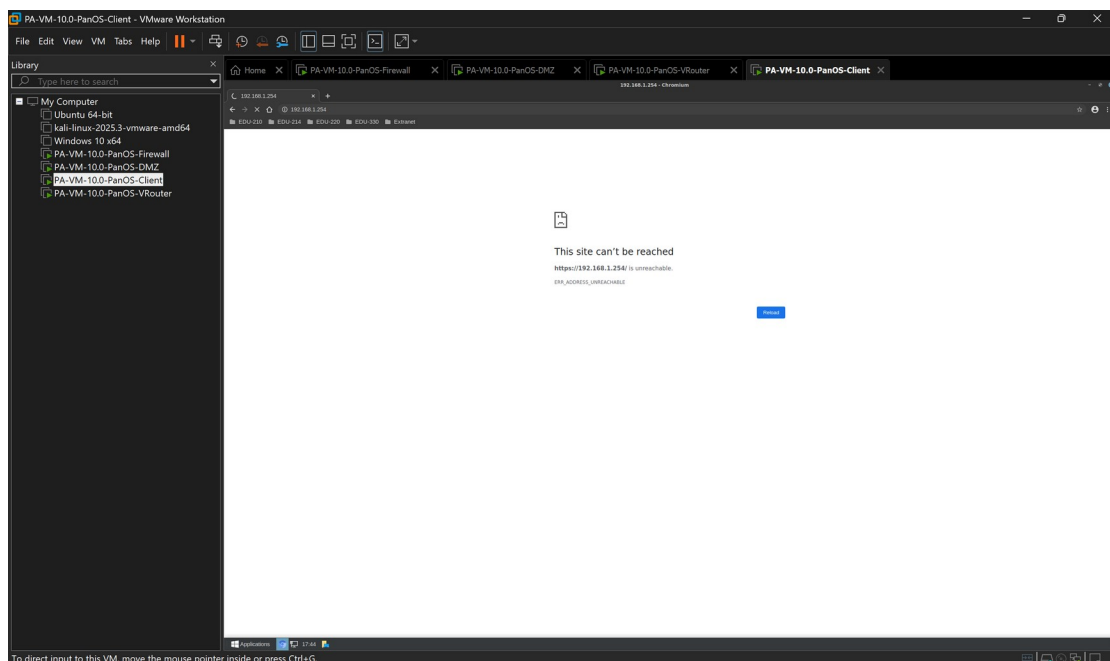
1. **Changing the Client's network:** First, we changed the settings of the Client virtual machine, moving the Network Adapter from the management network (VMnet1) to the internal Custom network (VMnet2). In this way, the Client now simulates an ordinary user located behind the Firewall rather than the system administrator.



2. **Internet Access Test (Deny-All):** We tried to connect to a random website (e.g. Google) through the browser. As shown in the relevant screenshot, the connection failed with a 'Server Not Found' message. This result is expected and desired. It confirms that the final 'Deny-All' rule works correctly, rejecting any traffic (such as internet access) that has not been explicitly permitted by the previous rules.



3. **Management Security Test:** Next, we tried to access the Firewall's management page again (192.168.1.254). The connection failed with the message 'This site can't be reached'. This confirms the correct isolation of the networks. The Client, now located in the internal network (Inside Zone), does not and should not have access to the Management Interface, thus protecting the Firewall from unauthorised interference.



In conclusion, the lab was completed successfully, as the Firewall allows only the specified services (SSH, Web, Port 9001 to the DMZ) and effectively blocks any other unauthorised traffic.

Bibliography

Shesh Chauhan IT Trainer. (2025, December 22). *Palo Alto Firewall Lab Guide | Step-by-Step Configuration Tutorial* | [Video file]. Retrieved from <https://www.youtube.com/watch?v=K2S7B4XDEXQ>

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